

# CH365 Chemical Engineering Thermodynamics

## Lesson 14

### Generalized Correlations for Gases and Liquids

# Generalized Correlations for Gases Slide 2

## Lee-Kesler Method

Byung Ik Lee and Michael Kesler,  
AIChE Journal, 1975, 21(3), 511-527

### Pitzer Correlation for Z (L13)

(Eq. 3.53)

$$Z = Z^0 + \omega Z^1$$

simple fluids

deviation from  
simple fluids:

$$\omega Z^1 = Z - Z^0$$

$$Z^0 = 1 + B^0 \frac{P_r}{T_r}$$

$$Z^1 = B^1 \cdot \frac{P_r}{T_r}$$

(Eq. 3.60)

Lesson 13, Slide 5  
formulas for  $B^0$  and  $B^1$   
eqns. 3.61 and 3.62

$$Z = 1 + \hat{B} \frac{P_r}{T_r} \text{ (eq. 3.57)}$$

$$\hat{B} = B^0 + \omega B^1 \text{ (eq. 3.59)}$$

$$\begin{aligned} Z &= 1 + (B^0 + \omega B^1) \frac{P_r}{T_r} \\ &= 1 + B^0 \frac{P_r}{T_r} + \omega B^1 \frac{P_r}{T_r} \\ &= Z^0 + \omega Z^1 \text{ (eq. 3.53, implied)} \end{aligned}$$

### Lee-Kesler Modification

$Z^{(r)}$ : calculated for n-octane

$$Z = Z^0 + \frac{\omega}{\omega^{(r)}} (Z^{(r)} - Z^0) \quad \text{where} \quad Z^1 = \frac{(Z^{(r)} - Z^0)}{\omega^{(r)}}$$

Lee and Kesler used a modified Benedict-Webb-Rubin EOS:

$$Z = 1 + \frac{B}{V_r} + \frac{C}{V_r^2} + \frac{D}{V_r^5} + \frac{c_4}{T_r^3 V_r^2} \left( \beta + \frac{\gamma}{V_r^2} \right) \exp \left( -\frac{\gamma}{V_r^2} \right)$$

$B, C, D$  are functions of  $T_r$   
(published in the paper)  
 $\beta, \gamma, c_4$ , etc. are constants

$Z$  calculated twice:

$Z^0$ : calculated for simple fluids

$Z^{(r)}$ : calculated for n-octane

Tables: Appendix – Tables D.1-D.4, pp. 676-692

Example: Find Z for n-octane at  $P_r=0.4$ ,  $T_r=0.9$

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Table D.1: Values of  $Z^0$  Page 677

| $P_r =$ | 0.0100 | 0.0500 | 0.1000 | 0.2000 | 0.4000 | 0.6000 | 0.8000 | 1.0000 |
|---------|--------|--------|--------|--------|--------|--------|--------|--------|
| $T_r$   |        |        |        |        |        |        |        |        |
| 0.30    | 0.0029 | 0.0145 | 0.0290 | 0.0579 | 0.1158 | 0.1737 | 0.2315 | 0.2892 |
| 0.35    | 0.0026 | 0.0130 | 0.0261 | 0.0522 | 0.1043 | 0.1564 | 0.2084 | 0.2604 |
| 0.40    | 0.0024 | 0.0119 | 0.0239 | 0.0477 | 0.0953 | 0.1429 | 0.1904 | 0.2379 |
| 0.45    | 0.0022 | 0.0110 | 0.0221 | 0.0442 | 0.0882 | 0.1322 | 0.1762 | 0.2200 |
| 0.50    | 0.0021 | 0.0103 | 0.0207 | 0.0413 | 0.0825 | 0.1236 | 0.1647 | 0.2056 |
| 0.55    | 0.9804 | 0.0098 | 0.0195 | 0.0390 | 0.0778 | 0.1166 | 0.1553 | 0.1939 |
| 0.60    | 0.9849 | 0.0093 | 0.0186 | 0.0371 | 0.0741 | 0.1109 | 0.1476 | 0.1842 |
| 0.65    | 0.9881 | 0.9377 | 0.0178 | 0.0356 | 0.0710 | 0.1063 | 0.1415 | 0.1765 |
| 0.70    | 0.9904 | 0.9504 | 0.8958 | 0.0344 | 0.0687 | 0.1027 | 0.1366 | 0.1703 |
| 0.75    | 0.9922 | 0.9598 | 0.9165 | 0.0336 | 0.0670 | 0.1001 | 0.1330 | 0.1656 |
| 0.80    | 0.9935 | 0.9669 | 0.9319 | 0.8539 | 0.0661 | 0.0985 | 0.1307 | 0.1626 |
| 0.85    | 0.9946 | 0.9725 | 0.9436 | 0.8810 | 0.0661 | 0.0983 | 0.1301 | 0.1614 |
| 0.90    | 0.9954 | 0.9768 | 0.9528 | 0.9015 | 0.7800 | 0.1006 | 0.1321 | 0.1630 |
| 0.93    | 0.9959 | 0.9790 | 0.9573 | 0.9115 | 0.8059 | 0.6635 | 0.1359 | 0.1664 |
| 0.95    | 0.9961 | 0.9803 | 0.9600 | 0.9174 | 0.8206 | 0.6967 | 0.1410 | 0.1705 |
| 0.97    | 0.9963 | 0.9815 | 0.9625 | 0.9227 | 0.8338 | 0.7240 | 0.5580 | 0.1779 |
| 0.98    | 0.9965 | 0.9821 | 0.9637 | 0.9253 | 0.8398 | 0.7360 | 0.5887 | 0.1844 |
| 0.99    |        |        |        |        |        |        |        |        |
| 1.00    |        |        |        |        |        |        |        |        |
| 1.01    |        |        |        |        |        |        |        |        |
| 1.02    |        |        |        |        |        |        |        |        |
| 1.05    |        |        |        |        |        |        |        |        |
| 1.10    |        |        |        |        |        |        |        |        |
| 1.15    |        |        |        |        |        |        |        |        |
| 1.20    |        |        |        |        |        |        |        |        |
| 1.30    |        |        |        |        |        |        |        |        |
| 1.40    |        |        |        |        |        |        |        |        |
| 1.50    |        |        |        |        |        |        |        |        |
| 1.60    |        |        |        |        |        |        |        |        |
| 1.70    |        |        |        |        |        |        |        |        |
| 1.80    |        |        |        |        |        |        |        |        |
| 1.90    |        |        |        |        |        |        |        |        |
| 2.00    |        |        |        |        |        |        |        |        |
| 2.20    |        |        |        |        |        |        |        |        |
| 2.40    |        |        |        |        |        |        |        |        |
| 2.60    |        |        |        |        |        |        |        |        |
| 2.80    |        |        |        |        |        |        |        |        |
| 3.00    |        |        |        |        |        |        |        |        |
| 3.50    |        |        |        |        |        |        |        |        |
| 4.00    |        |        |        |        |        |        |        |        |

$Z^0=0.780$

$Z^1=-0.1118$

$\omega=0.400$

$$Z = Z^0 + \omega Z^1$$

$$= 0.780 + (0.400) \cdot (-0.1118)$$

$$= .7353$$

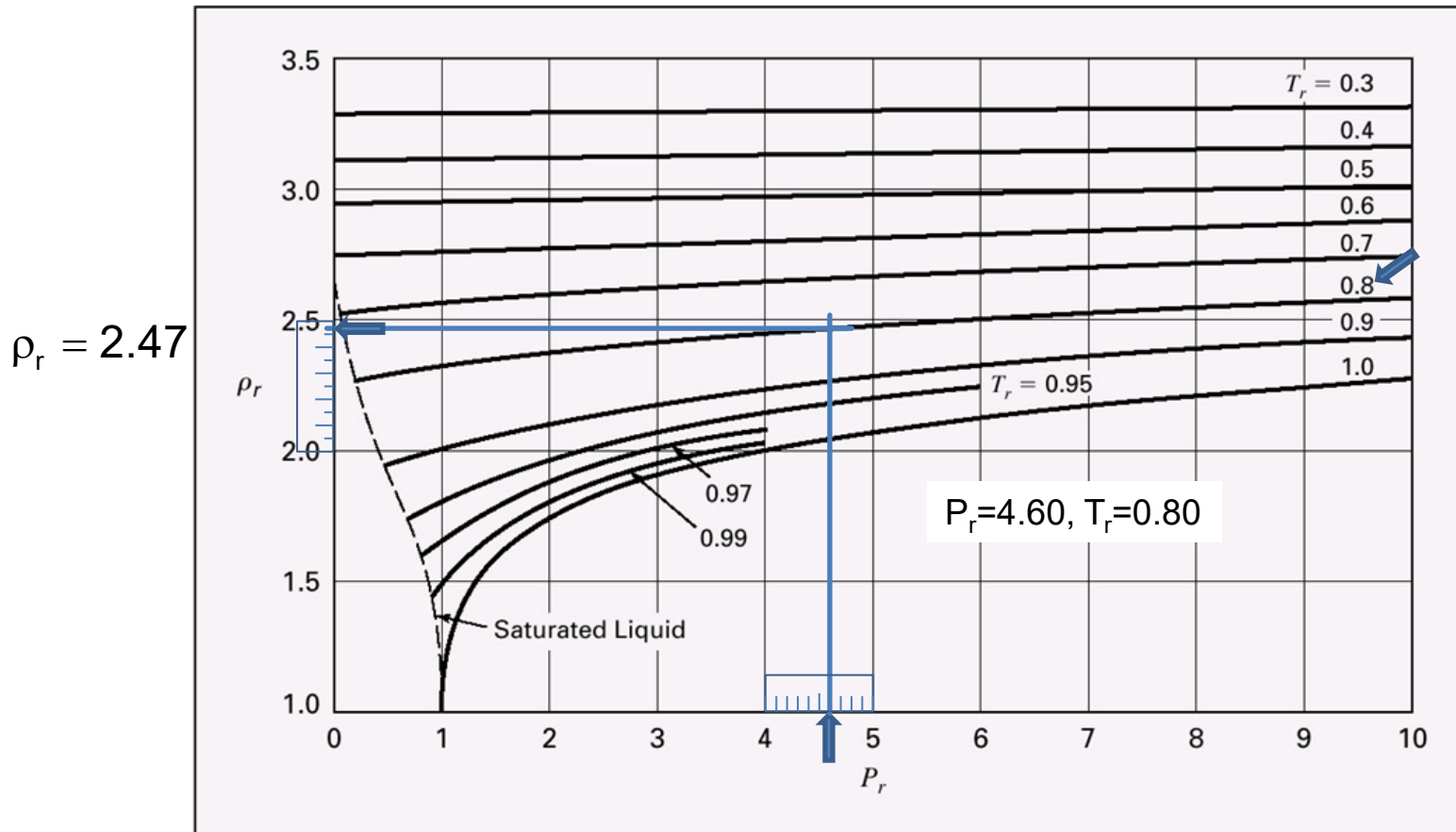
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Table D.2: Values of  $Z^1$  Page 678

| $P_r =$ | 0.0100  | 0.0500  | 0.1000  | 0.2000  | 0.4000  | 0.6000  | 0.8000  | 1.0000  |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| $T_r$   |         |         |         |         |         |         |         |         |
| 0.30    | -0.0008 | -0.0040 | -0.0081 | -0.0161 | -0.0323 | -0.0484 | -0.0645 | -0.0806 |
| 0.35    | -0.0009 | -0.0046 | -0.0093 | -0.0185 | -0.0370 | -0.0554 | -0.0738 | -0.0921 |
| 0.40    | -0.0010 | -0.0048 | -0.0095 | -0.0190 | -0.0380 | -0.0570 | -0.0758 | -0.0946 |
| 0.45    | -0.0009 | -0.0047 | -0.0094 | -0.0187 | -0.0374 | -0.0560 | -0.0745 | -0.0929 |
| 0.50    | -0.0009 | -0.0045 | -0.0090 | -0.0181 | -0.0360 | -0.0539 | -0.0716 | -0.0893 |
| 0.55    | -0.0314 | -0.0043 | -0.0086 | -0.0172 | -0.0343 | -0.0513 | -0.0682 | -0.0849 |
| 0.60    | -0.0205 | -0.0041 | -0.0082 | -0.0164 | -0.0326 | -0.0487 | -0.0646 | -0.0803 |
| 0.65    | -0.0137 | -0.0772 | -0.0078 | -0.0156 | -0.0309 | -0.0461 | -0.0611 | -0.0759 |
| 0.70    | -0.0093 | -0.0507 | -0.1161 | -0.0148 | -0.0294 | -0.0438 | -0.0579 | -0.0718 |
| 0.75    | -0.0064 | -0.0339 | -0.0744 | -0.0143 | -0.0282 | -0.0417 | -0.0550 | -0.0681 |
| 0.80    | -0.0044 | -0.0228 | -0.0487 | -0.1160 | -0.0272 | -0.0401 | -0.0526 | -0.0648 |
| 0.85    | -0.0029 | -0.0152 | -0.0319 | -0.0715 | -0.0268 | -0.0391 | -0.0509 | -0.0622 |
| 0.90    | -0.0019 | -0.0099 | -0.0205 | -0.0442 | -0.1118 | -0.0396 | -0.0503 | -0.0604 |
| 0.93    | -0.0015 | -0.0075 | -0.0154 | -0.0326 | -0.0763 | -0.1662 | -0.0514 | -0.0602 |
| 0.95    | -0.0012 | -0.0062 | -0.0126 | -0.0262 | -0.0589 | -0.1110 | -0.0540 | -0.0607 |
| 0.97    | -0.0010 | -0.0050 | -0.0101 | -0.0208 | -0.0450 | -0.0770 | -0.1647 | -0.0623 |
| 0.98    | -0.0009 | -0.0044 | -0.0090 | -0.0184 | -0.0390 | -0.0641 | -0.1100 | -0.0641 |
| 99      | -0.0008 | -0.0039 | -0.0079 | -0.0161 | -0.0335 | -0.0531 | -0.0796 | -0.0680 |
| 10      | -0.0007 | -0.0034 | -0.0069 | -0.0140 | -0.0285 | -0.0435 | -0.0588 | -0.0879 |
| 11      | -0.0006 | -0.0030 | -0.0060 | -0.0120 | -0.0240 | -0.0351 | -0.0429 | -0.0223 |
| 12      | -0.0005 | -0.0026 | -0.0051 | -0.0102 | -0.0198 | -0.0277 | -0.0303 | -0.0062 |
| 15      | -0.0003 | -0.0015 | -0.0029 | -0.0054 | -0.0092 | -0.0097 | -0.0032 | 0.0220  |
| 10      | 0.0000  | 0.0000  | 0.0001  | 0.0007  | 0.0038  | 0.0106  | 0.0236  | 0.0476  |
| 15      | 0.0002  | 0.0011  | 0.0023  | 0.0052  | 0.0127  | 0.0237  | 0.0396  | 0.0625  |
| 20      | 0.0004  | 0.0019  | 0.0039  | 0.0084  | 0.0190  | 0.0326  | 0.0499  | 0.0719  |
| 30      | 0.0006  | 0.0030  | 0.0061  | 0.0125  | 0.0267  | 0.0429  | 0.0612  | 0.0819  |
| 40      | 0.0007  | 0.0036  | 0.0072  | 0.0147  | 0.0306  | 0.0477  | 0.0661  | 0.0857  |
| 50      | 0.0008  | 0.0039  | 0.0078  | 0.0158  | 0.0323  | 0.0497  | 0.0677  | 0.0864  |
| 50      | 0.0008  | 0.0040  | 0.0080  | 0.0162  | 0.0330  | 0.0501  | 0.0677  | 0.0855  |
| 70      | 0.0008  | 0.0040  | 0.0081  | 0.0163  | 0.0329  | 0.0497  | 0.0667  | 0.0838  |
| 30      | 0.0008  | 0.0040  | 0.0081  | 0.0162  | 0.0325  | 0.0488  | 0.0652  | 0.0814  |
| 30      | 0.0008  | 0.0040  | 0.0079  | 0.0159  | 0.0318  | 0.0477  | 0.0635  | 0.0792  |
| 30      | 0.0008  | 0.0039  | 0.0078  | 0.0155  | 0.0310  | 0.0464  | 0.0617  | 0.0767  |
| 20      | 0.0007  | 0.0037  | 0.0074  | 0.0147  | 0.0293  | 0.0437  | 0.0579  | 0.0719  |
| 40      | 0.0007  | 0.0035  | 0.0070  | 0.0139  | 0.0276  | 0.0411  | 0.0544  | 0.0675  |
| 50      | 0.0007  | 0.0033  | 0.0066  | 0.0131  | 0.0260  | 0.0387  | 0.0512  | 0.0634  |
| 30      | 0.0006  | 0.0031  | 0.0062  | 0.0124  | 0.0245  | 0.0365  | 0.0483  | 0.0598  |
| 30      | 0.0006  | 0.0029  | 0.0059  | 0.0117  | 0.0232  | 0.0345  | 0.0456  | 0.0565  |
| 50      | 0.0005  | 0.0026  | 0.0052  | 0.0103  | 0.0204  | 0.0303  | 0.0401  | 0.0497  |
| 4.00    | 0.0005  | 0.0023  | 0.0046  | 0.0091  | 0.0182  | 0.0270  | 0.0357  | 0.0443  |

# Generalized Correlations for Liquids Slide 4

Rackett:  $V^{\text{sat}} = V_C Z_C^{(1-T_r)^{2/7}}$  (Eq. 3.68)  $Z^{\text{sat}} = \frac{P_r}{T_r} Z_C^{[1+(1-T_r)^{2/7}]}$  (Eq. 3.69)



**Figure 3.15:** Generalized density correlation for liquids.

Lydersen, Greenkorn, and Hougen:  $\rho_r \equiv \frac{\rho}{\rho_C} = \frac{V_C}{V}$  (Eq. 3.70)

# Example 3.14

For ammonia at 310 K, estimate the molar volume density of (a) the saturated liquid and (b) the liquid at 100 bar.

## (Liquids)

## Example 3.14 , part b, continued

Rackett:  $V^{\text{sat}} = V_C Z_C^{(1-T_r)^{2/7}}$  (Eq. 3.68)  $Z^{\text{sat}} = \frac{P_r}{T_r} Z_C^{[1+(1-T_r)^{2/7}]}$  (Eq. 3.69)

$$\rho_r = 2.43$$

solution deviates  
from book;  
authors read  
reduced density  
as 2.38 on p. 117

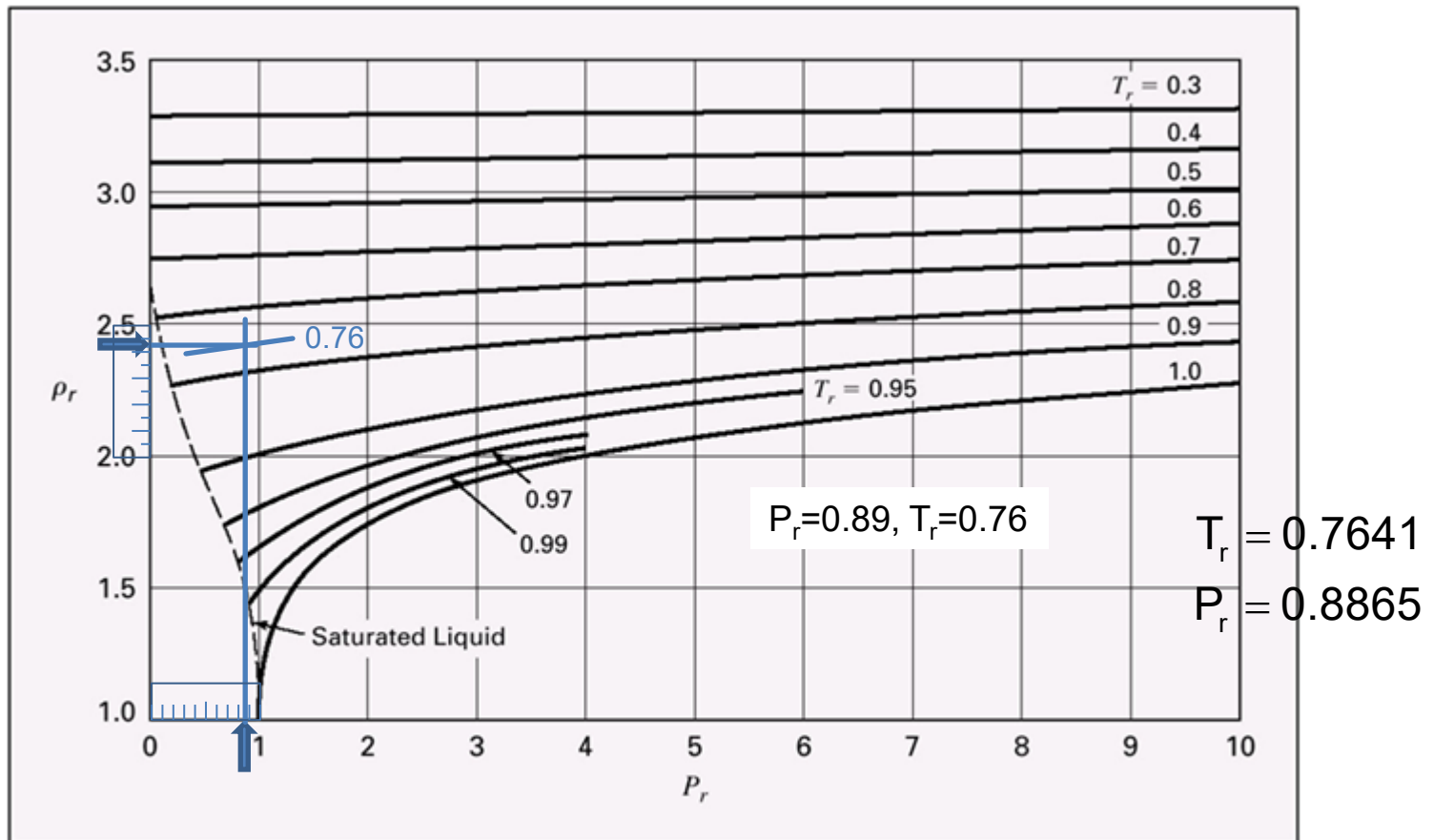


Figure 3.15. Generalized density correlation for liquids.

Lydersen, Greenkorn, and Hougen:  $\rho_r \equiv \frac{\rho}{\rho_C} = \frac{V_C}{V}$  (Eq. 3.70)

## Example 3.14, continued

For ammonia at 310 K, estimate the molar volume density of (a) the saturated liquid and (b) the liquid at 100 bar.

# Questions

# Homework

# Problem 3.58

To a good approximation, what is the molar volume of ethanol vapor at 480 deg C and 6000 kPa? How does this result compare with the ideal gas?

Answer the problem in four parts:

(a) Lee-Kesler method.

(b) SRK equation.

(c) Ideal gas equation.

For comparison: Compare LK and SRK to IG. If either is less than IG, explain why using knowledge of IG behavior.

Online Interpolator Tool for Lee-Kesler Tables:

[https://www.ajdesigner.com/phpinterpolation/bilinear\\_interpolation\\_equation.php](https://www.ajdesigner.com/phpinterpolation/bilinear_interpolation_equation.php)